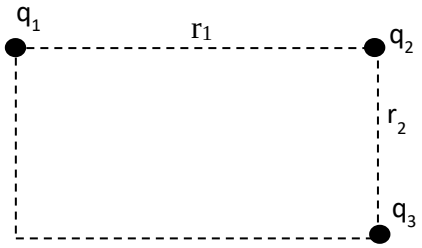
Physics 2 Problem Solving Video # 2 – Electro-statics

Shown in the figure below, three charged objects are placed at corners of a rectangle. The magnitude and type of charge is given below. Determine the magnitude and direction of the net electrostatic force acting on object 1. (k = 9.0 x 10^9 Nm^2/C^2)

q_1 = 10 μC and is positive.
q_2 = 5 μC and is positive.
q_3 = 15 μC and is negative.

r_1 = 0.5 m
r_2 = 0.4 m



Net force is equal to $\Sigma\vec{F} = \vec{F}_{2\ on\ 1} + \vec{F}_{3\ on\ 1}$

Answer: $\left|\Sigma\vec{F}\right|$ = 2.2 N α = 69° and β = 159°